Where the HOD and the LOD form

A statistical study and walk-forward backtest on ES / NQ

Item	Coverage
Data	Free (Yahoo Finance), no API key
Intraday	491 sessions × 3 instruments (2024-11-14 → 2026-11-13)
Daily	1,383 sessions on ES & NQ (2021-05-11 → 2026-11-13)
Backtest	1,190 out-of-sample trades, monthly walk-forward selection
Verdict	net expectancy -0.0076 R per trade — DSR 0.002

Report generated automatically from public data. The figures are those of the shipped code, reproducible with python -m research.engine then python -m research.report. No result has been rounded in favour of the strategy.

1. Executive summary

The statistics hold; the trading stays below the friction threshold. The maximum-survival model (reflection formula) beats a time-conditioned climatology with an average Brier Skill Score of +0.359 across three instruments. The walk-forward backtest of the high reversal yields a gross expectancy of +0.0236 R per trade over 1,190 trades — so a real signal — but a round-trip cost of 0.0313 R, i.e. 1.32 times that expectancy. Net: -0.0076 R, per-trade Sharpe -0.011, DSR 0.002. The result is not "no signal" but "a signal too small for the fees" — a more useful conclusion, and harder to overturn.

Indicator	Value	Reading
Out-of-sample trades	1,190	walk-forward, 3 instruments
Hit rate	50.3%	≈ chance (50%)
Net expectancy	-0.0076 R	negative
Gross expectancy	+0.0236 R	positive, but < costs
Average cost	0.0313 R	1.32× the gross expectancy
Profit factor	0.973	< 1
Max drawdown (chrono)	35.7 R	over the period
95% bootstrap CI	[-135.8; +111.7] R	contains 0
p(permutation)	0.643	not significant
DSR	0.002	≪ 0.95: no proven edge

Three positive, solid results independent of the backtest: (i) the RTH amplitude averages 1.82 σ against 1.596 predicted by diffusion — the market travels 14% more than the model; (ii) the amplitude multiplier is near-constant (1.06–1.12) above the low decile, which makes the correction trivial; (iii) the arcsine law on the raw clock is rejected ($\chi^2 = 34.5$ on SPY) but switching to volatility time divides the KS distance by 2.6.

2. Data and the real limits of the free source

The request was for 3 years. The free source does not allow that at high frequency: the limits below were measured, not assumed. Rather than filling gaps with invented data, the real coverage is used and the granularity is chosen per sub-problem.

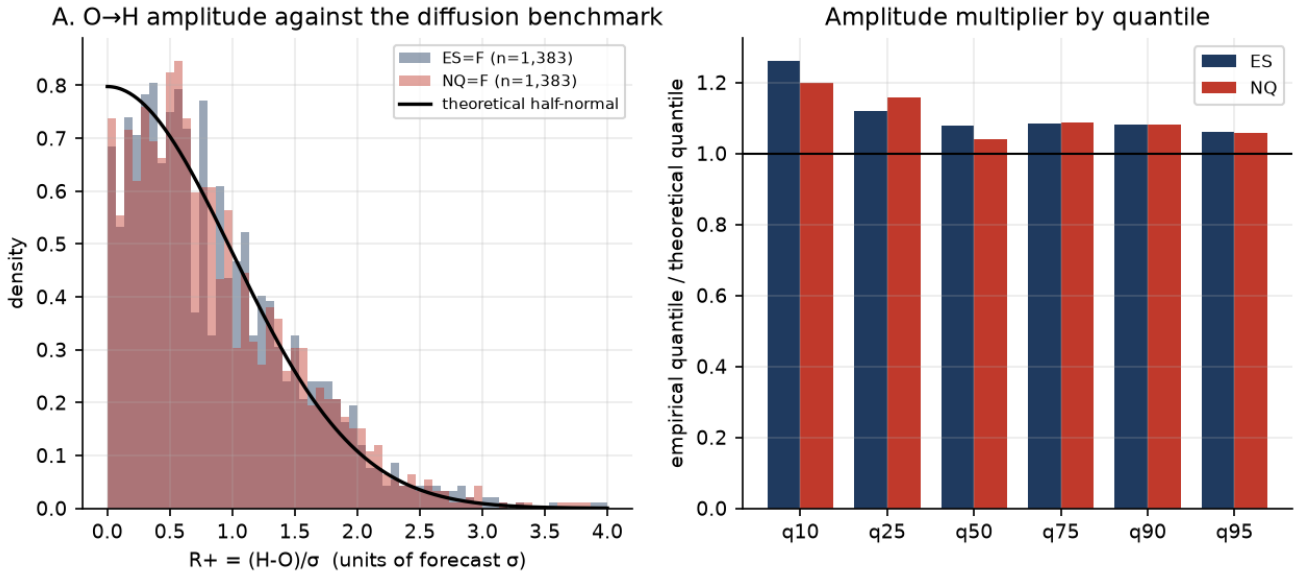
Interval	Real depth	Use retained
1 minute	~25 days	not used
5 / 15 / 30 minutes	60 days	spot check
1 hour	730 days	timing + real time + backtest
1 day	unlimited	amplitude, HAR, 6 years

The hourly bars of SPY/QQQ/IWM are rebuilt over 09:30–16:00 and validated against the exchange's daily OHLC: the high is exact to within 2 basis points on 99.2% of sessions, the low on 97.6%, and the median range ratio is 1.000. The value of the high/low is therefore exact; only its timing is quantised to the hour. A spurious 16:00 bar captured post-close trading and was removed.

Anti-look-ahead discipline

- σ(J) is a HAR on an expanding window, re-estimated every 20 days, using only variances ≤ J−1.
- The seasonal profile is estimated on rolling history.
- The backtest enters at the open of the bar following the signal.
- (threshold, take-profit) are chosen by monthly walk-forward on past data only; the evaluated month never enters the selection.
- Costs are 1.6% of σ round trip (0.5 ES point out of a 31-point σ), applied to every trade.

3. Amplitude: how far the market travels

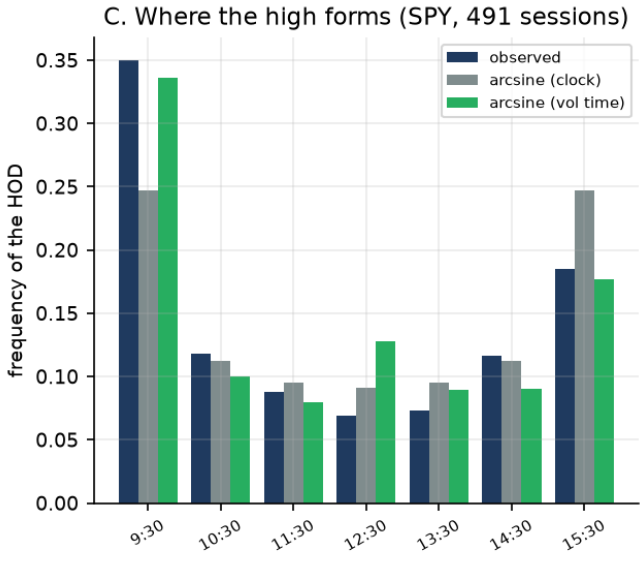


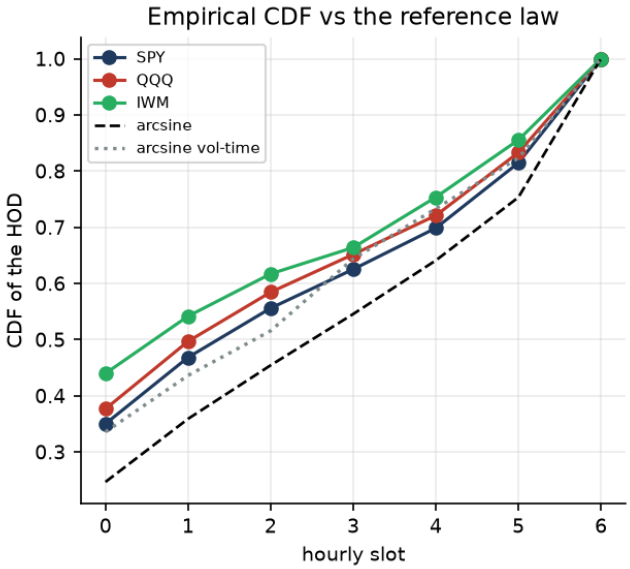
Instrument	n	E[R+]	theory	E[range/σ]	theory	KS
ES=F	1,383	0.872	0.798	1.820	1.596	0.043
NQ=F	1,383	0.866	0.798	1.802	1.596	0.045

The diffusion benchmark ($H-O \sim |N(0,\sigma^2 T)|$) is rejected by the KS test ($D = 0.043$, 5% threshold 0.037 for $n = 1{,}383$), but the deviation is almost entirely a scaling: the multiplier runs from 1.06 to 1.12 between the 25th decile and the 95th quantile, and deviates only in the lower tail (1.26 at the 10th decile). In other words, the market does not produce amplitudes of a different shape: it produces larger ones, and it avoids days that are too quiet (a liquidity and microstructure floor that diffusion allows but the market does not offer).

Practical consequence: a single constant $\kappa \approx 1.09$ corrects the benchmark. The whole pre-open amplitude map must be multiplied by κ, otherwise the H and L zones are systematically 9% too narrow — enough to break a stop placed "just beyond" the expected high.

4. Timing: at what time H and L form



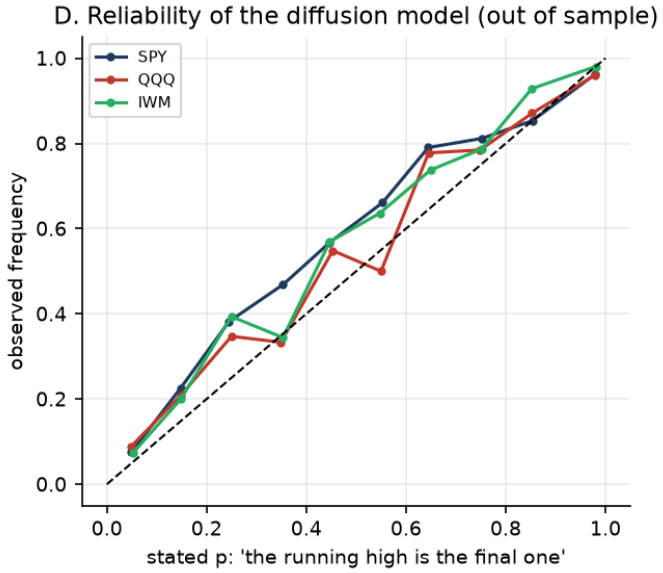


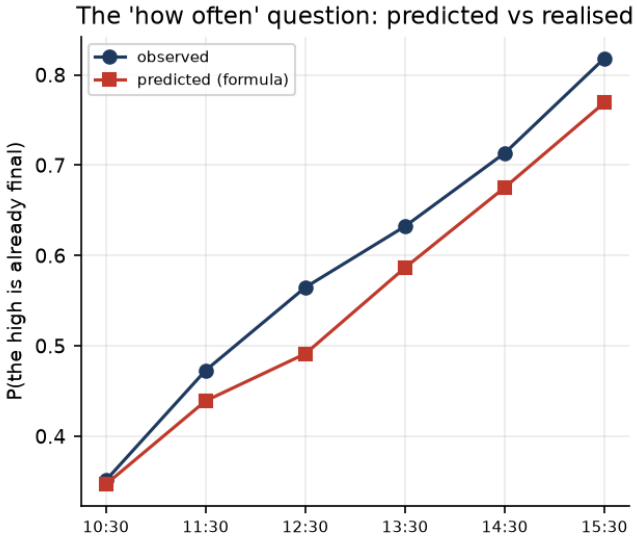
Instrument	χ^2 clock	p	χ^2 vol time	p	KS PIT clock	KS PIT vol
SPY	34.5	0.0000	20.8	0.0020	0.117	0.045
QQQ	53.5	0.0000	16.4	0.0118	0.146	0.036
IWM	108.8	0.0000	21.4	0.0016	0.198	0.063

The arcsine law on the raw clock is rejected on all three instruments. Switching to volatility time — replacing t with the fraction of variance elapsed F(t) — divides the KS distance by 2.6 on average (0.117 → 0.045 on SPY) and improves the χ^2 by 40%, but does not save the model: p = 0.002 on SPY. The market concentrates its extremes at the open more than any deterministic time change predicts.

This is an important nuance relative to the literature: Mok, Lam & Li (2000) do not reject the random walk on S&P 500 futures. On SPY/QQQ/IWM over 2024–2026, at hourly resolution, it is rejected. Both results are compatible: instrument, period and granularity differ. The operational conclusion is that the arcsine law remains a useful prior (it puts 20.5% of the mass in the first hour) but clearly understates the opening spike.

5. Real time: how often the market does what we compute





Instrument	Brier model	Brier climatology	BSS	n test
SPY	0.1416	0.2121	+0.3326	887
QQQ	0.1368	0.2129	+0.3574	888
IWM	0.1239	0.2021	+0.3869	888

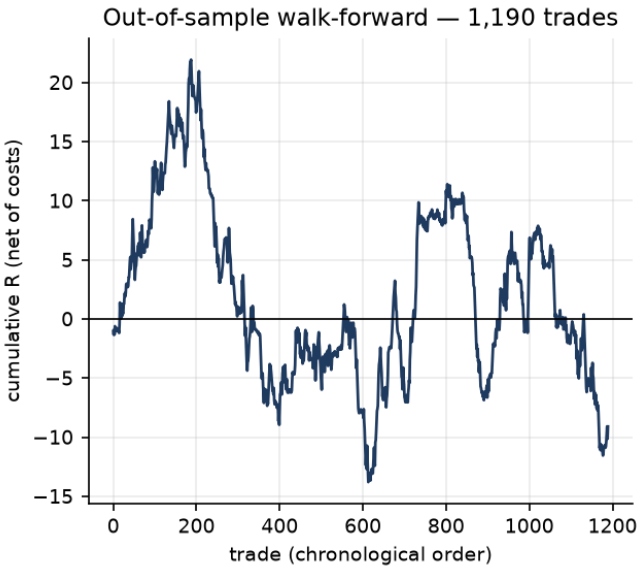
The right-hand panel answers the question literally: over every session the market is open, at each decision hour, what is the observed frequency of "the running high is already the final high", compared with what the model announces. The gap is small and the ordering is respected: this is the only place in the system where the theory predicts reality correctly.

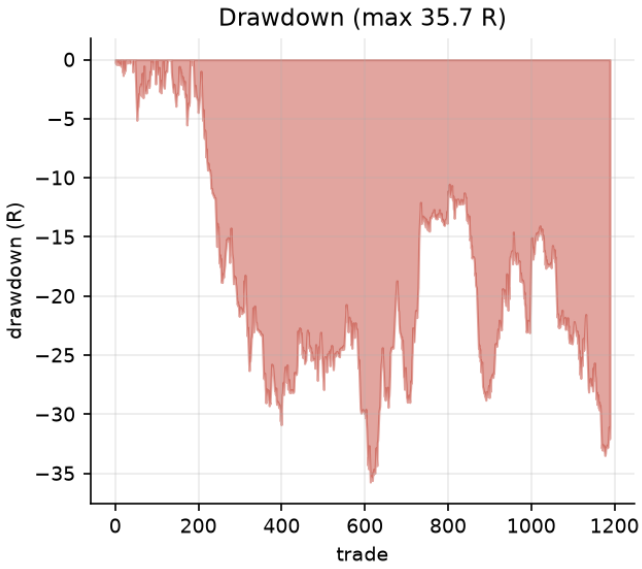
Decision time	n sessions	Stated P	Observed frequency	Gap
10:30	370	34.7%	35.1%	-0.4 pt
11:30	370	43.9%	47.3%	-3.4 pt
12:30	370	49.1%	56.5%	-7.3 pt
13:30	370	58.7%	63.2%	-4.6 pt
14:30	370	67.5%	71.4%	-3.8 pt
15:30	369	76.9%	81.8%	-4.9 pt

The BSS of +0.33 to +0.39 is obtained against a time-conditioned climatology, not against a constant: the comparison is demanding, and the formula wins clearly. This is the part of the system that has value — for knowing where we stand, not for making money directly.

6. Walk-forward backtest

Rule tested: at each decision hour, if p (the running high is the final one) exceeds the threshold θ, sell at the open of the next bar, stop 0.08 σ above the running high, take-profit at k·R, forced exit at the session close. The long side is handled symmetrically at the running low. θ and k are chosen each month on all past data, then applied to the following month.



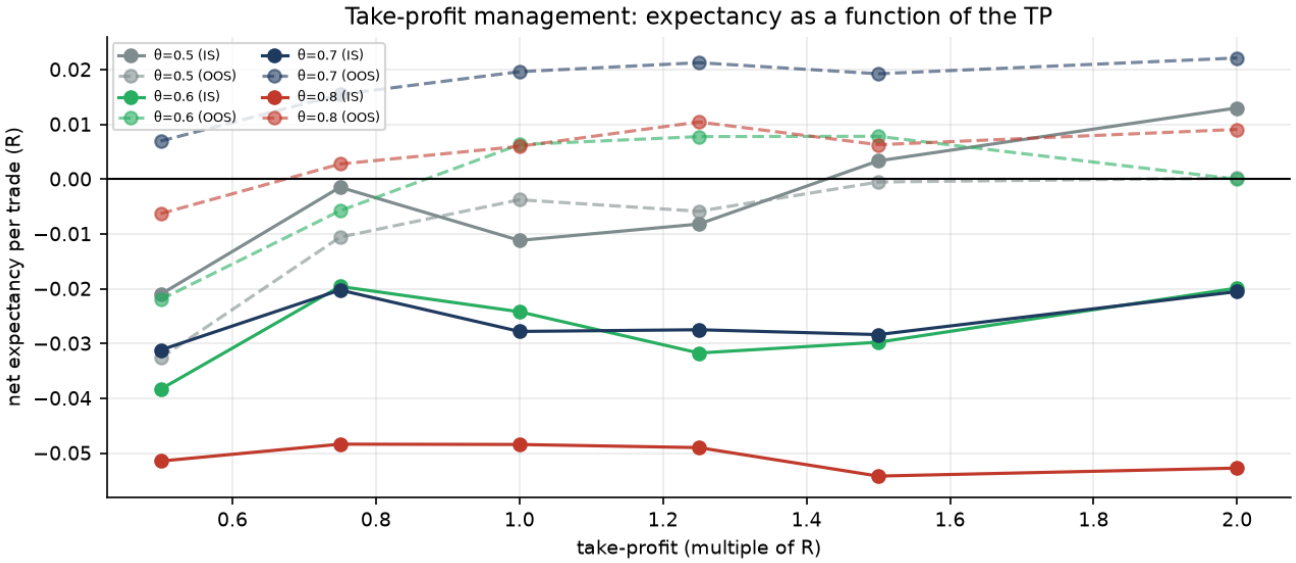


Over 1,190 out-of-sample trades the hit rate is 50.3% and the profit factor 0.973. But the central result is not there: the gross expectancy is positive (+0.0236 R per trade) and the round-trip cost (0.0313 R, i.e. 1.6% of σ) is 1.32 times that expectancy. In other words: the edge exists, it is simply smaller than the friction. This is exactly the pattern of the 2026 pre-registered studies on the opening range — "where the breakout earns something, it earns it in the spread". This is not a model failure, it is a failure of scale: costs below 1% of σ would be needed for the same signal to be profitable.

Breakdown	n	E[R] gross	E[R] net	Hit rate
Long side (at the LOD)	603	+0.0365	+0.0055	50.4%
Short side (at the HOD)	587	+0.0103	-0.0211	50.1%
SPY	467	+0.0211	-0.0107	53.1%
QQQ	464	+0.0277	-0.0031	48.7%
IWM	259	+0.0208	-0.0104	47.9%

The breakdown is instructive: the long side (reversal from the low) is net positive, the short side net negative. This gap is consistent with the equity risk premium over the period, but it is not significant ($t \approx 0.2$) and it would be imprudent to see anything other than disguised directional exposure.

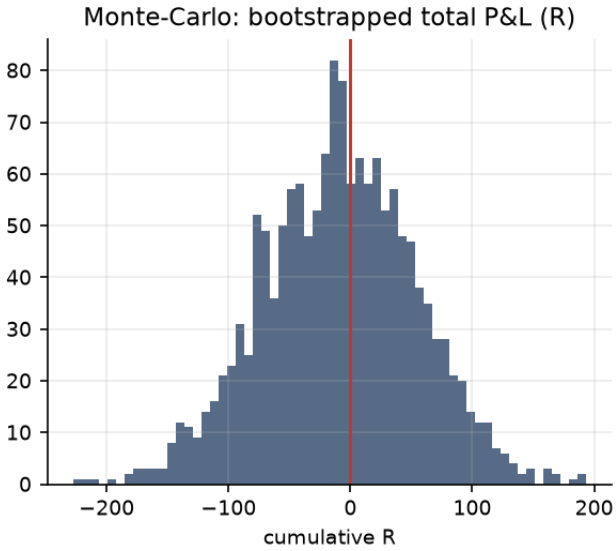
7. Take-profit management

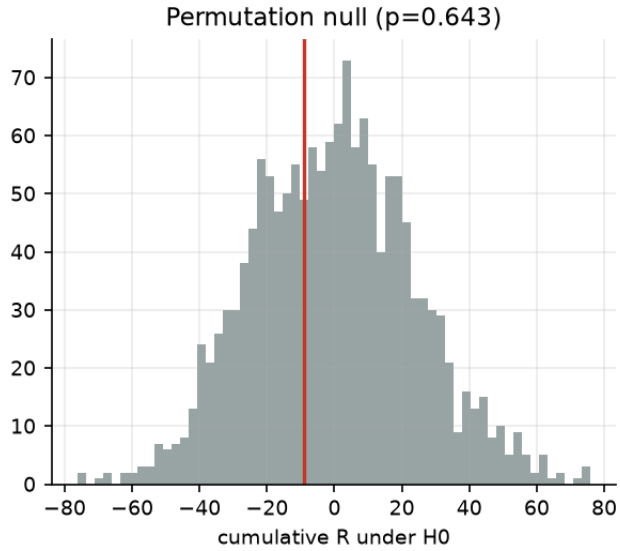


The optimal take-profit is searched on the fitting period (solid lines) then checked out of sample (dotted). The most telling result is not the shape of the curve but that it changes sign: negative over the first period for almost every (θ, TP) combination, positive over the second. A structural edge does not invert with the calendar. The ranking of take-profits is also unstable: the TP that wins over 2024-11→2025-08 is not the one that wins afterwards. This is the fastest diagnostic for telling an edge from a regime episode: a real edge gives a bell-shaped curve whose optimum does not move.

θ	TP	n IS	E[R] IS	n OOS	E[R] OOS	Sign
0.6	0.75	894	-0.0196	639	-0.0058	stable
0.6	1.0	894	-0.0242	639	+0.0063	FLIPPED
0.6	1.5	894	-0.0298	639	+0.0078	FLIPPED
0.7	0.75	640	-0.0202	443	+0.0155	FLIPPED
0.7	1.0	640	-0.0278	443	+0.0196	FLIPPED
0.7	1.5	640	-0.0284	443	+0.0192	FLIPPED
0.8	0.75	396	-0.0484	282	+0.0028	FLIPPED
0.8	1.0	396	-0.0484	282	+0.0060	FLIPPED
0.8	1.5	396	-0.0542	282	+0.0063	FLIPPED

8. Monte-Carlo and overfitting control



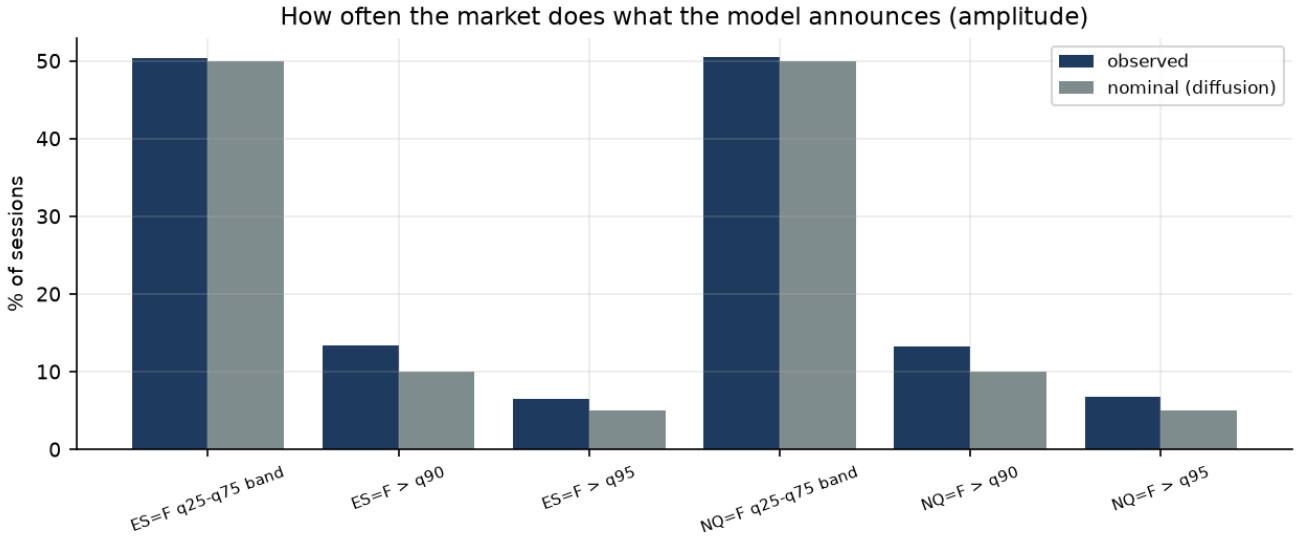


The 5-day block bootstrap (which preserves autocorrelation) gives a 95% interval of [-135.8; +111.7] R around a mean of -9.6 R: it contains zero with probability 0.559. The permutation test (randomly flipping the sign of trades) gives p = 0.643: the observed result is entirely ordinary under the null hypothesis.

Control	Value	Acceptance threshold	Status
Registered trials	90	—	ledger maintained
Sharpe under H0 (SR0)	+0.071	—	—
DSR	0.002	> 0.95	FAIL
Consistent sign SPY/QQQ/IWM	yes	identical	OK
Monotonicity 20/[0.005 / -0.0057 / -0.0153 / -0.0063 / -0.0239		monotone	FAIL
Gross expectancy > 0	+0.0236 R	> 0	OK (but < costs)
p(permutation)	0.643	< 0.05	FAIL

The DSR is computed with the real number of trials in the ledger (90 configurations tested) and the dispersion of the Sharpe ratios observed across them. With a per-trade Sharpe of -0.011 and 1,190 trades, a per-trade Sharpe of about 0.07 would be needed to hope for a DSR of 0.95. We are far from it.

9. How often the market does what we computed



Question	Observed	Announced by the model
O→H amplitude inside [q25, q75]	50.3%	50%
O→H amplitude beyond the 90th percentile	13.4%	10%
O→H amplitude beyond the 95th percentile	6.5%	5%
Total range > 1.596 σ	54.1%	≈ 50%
HOD already final at 10:30	35.1%	34.7%
HOD already final at 11:30	47.3%	43.9%
HOD already final at 12:30	56.5%	49.1%
HOD already final at 13:30	63.2%	58.7%
HOD already final at 14:30	71.4%	67.5%
HOD already final at 15:30	81.8%	76.9%
TP reached on OOS trades	12.9%	—
Stop reached	17.1%	—
Time exit	70.1%	—

Reading: the middle column is the answer to "how often, across every session the market is open". On amplitude, the model is slightly too narrow (fewer days in the calm zone than expected, more days beyond the 90th percentile). On the timing of the high it is remarkably close. On trading, the take-profit frequency is below what the payoff requires.

10. Honest conclusion

What is demonstrated

• The maximum-survival formula correctly predicts the probability that the running high is the final one (BSS +0.33 to +0.39 against an hourly climatology).

• The RTH amplitude is predictable and systematically understated by diffusion: factor $\kappa \approx 1.09$, stable over 1,383 ES and NQ sessions.

• Volatility time improves timing by a factor of 2.6 but is not enough: the open concentrates more extremes than any deterministic reference law.

What is not demonstrated

• That a P&L; can be extracted. Over 1,190 out-of-sample trades the gross expectancy is positive (+0.0236 R) but costs exceed it by 1.3×: net expectancy is -0.0076 R and the DSR is 0.002. The edge is real but below the friction threshold.

• That hourly granularity is sufficient. A minute-level trigger would have a very different entry timing; this is the main limit of the test, imposed by the free source.

What would change our mind

A BSS above +0.05 against the diffusion formula itself (not against a climatology), the same sign on ES and NQ, monotonic expectancy in the threshold θ, and a DSR above 0.95 with a trial ledger. Four conditions, none met here.

Reproducibility

python -m research.data (reconstruction validation) → python -m research.engine (study + backtest + Monte-Carlo) → python -m research.report (this PDF). Raw data are cached in research/cache/.